

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	I.JASWANTH	Reg. No.:	192525345
Section / Batch:	AI AND ML	Date:	24/07/2026

Code of Conduct

I I.JASWANTH(192525345) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: I.JASWANTH

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25

Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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▼

Q7

Stack

5 marks

✓

Q8

Linked List

5 marks

✓

Q9

Linked List

5 marks

✓

Q10

Queue

5 marks

✓

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int key) {
4     int low = 0, high = n - 1;
5
6     while (low <= high) {
7         int mid = (low + high) / 2;
8
9         if (arr[mid] == key)
10            return mid;
11        else if (arr[mid] < key)
12            low = mid + 1;
13        else
14            high = mid - 1;
15    }
16
17    return -1;
18 }
19
20 int main() {
21     int arr[] = {10, 20, 30, 40, 50};
22     int n = 5, key = 30;
23 }
```

Test Input

10 20 30 40 50
30

Output

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▼

Q5

Stack - Balancing Symbols

5 marks

✓

Q6

Singly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List

5 marks

✓

Q9

Linked List

5 marks

✓

Q10

Queue

5 marks

✓

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11
12    return stack[top--];
13 }
14
15 int main() {
16     int x = pop();
17
18     if (x != -1)
19         printf("Popped element = %d", x);
20
21     return 0;
22 }
```

Test Input

{10, 20, 30}
2

Output

Popped element = 30

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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = -1;
8 }
9
10 int main() {
11     init();
12     printf("Top = %d", top);
13     return 0;
14 }
```

Submitted

Test Input

stdin input (optional)

Output

Top = -1

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (ch == '(') {
2     while (stack[top] != '(') {
3         postfix[j++] = pop();
4     }
5     pop();
6 }
```

Test Input

(A+B)*C

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192525345
RA_6a7982378893b7.19692375\main.c
:1:1: error: expected identifier
or '(' before 'if'
1 | if (ch == '(') {

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Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 char stack[10] = {' '};
4 int top = 0;
5
6 void pop() {
7     top--;
8 }
9
10 int main() {
11     char ch = ' ';
12
13     if ((ch == '[' && stack[top] == '[') ||
14         (ch == ')' && stack[top] == '(')) {
15         pop();
16         printf("Balanced Brackets");
17     } else {
18         printf("Not Balanced");
19     }
20
21     return 0;
22 }
```

Test Input

```
{}
```

Output

Balanced Brackets

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Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *second, *third, *temp;
11
12     head = (struct Node *)malloc(sizeof(struct Node));
13     second = (struct Node *)malloc(sizeof(struct Node));
14     third = (struct Node *)malloc(sizeof(struct Node));
15
16     head->data = 10;
17     head->next = second;
18
19     second->data = 20;
20     second->next = third;
21
22     third->data = 30;
23     third->next = NULL;
24
25     temp = head;
```

10 -> 20 -> 30 -> NULL

Output

10 20 30

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Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    } else {
11        return stack[top--];
12    }
13 }
14
15 int main() {
16     int item = pop();
17
18     if (item != -1)
19         printf("Popped element = %d", item);
20
21     return 0;
22 }
```

Test Input

{10, 20, 30}
2

Output

Popped element = 30

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Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *second, *third;
11     struct Node *current, *prev, *next;
12
13     head = (struct Node *)malloc(sizeof(struct Node));
14     second = (struct Node *)malloc(sizeof(struct Node));
15     third = (struct Node *)malloc(sizeof(struct Node));
16
17     head->data = 10;
18     head->next = second;
19
20     second->data = 20;
21     second->next = third;
22
23     third->data = 30;
24     third->next = NULL;
```

Test Input

10 -> 20 -> 30 -> NULL

Output

Reversed List: 30 20 10

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *newNode;
12
13     newNode = (struct Node *)malloc(sizeof(struct Node));
14
15     newNode->data = 10;
16     newNode->next = head;
17     head = newNode;
18
19     printf("Inserted element = %d\n", head->data);
20
21     return 0;
22 }
```

Test Input

10

Output

Inserted element = 10

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Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int Q[MAX] = {10, 20, 30};
6 int front = 0, rear = 2;
7
8 int dequeue() {
9     if (front == -1) {
10         printf("Underflow\n");
11         return -1;
12     }
13
14     int item = Q[front++];
15
16     if (front > rear) {
17         front = rear = -1;
18     }
19
20     return item;
21 }
22
23 int main() {
```

Test Input

{10, 20, 30}
0
2

Output

Deleted element = 10

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int peek() {
7     if (top == -1) {
8         printf("Stack Underflow\n");
9         return -1;
10    }
11    return stack[top];
12 }
13
14 int main() {
15     int item = peek();
16
17     if (item != -1)
18         printf("Top element = %d", item);
19
20     return 0;
21 }
```

Test Input

{10, 20, 30}
2

Output

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Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = 0, rear = 2;
7
8 void enqueue(int vehicle) {
9     if ((rear + 1) % MAX == front) {
10         printf("Queue Full");
11         return;
12     }
13
14     rear = (rear + 1) % MAX;
15     queue[rear] = vehicle;
16
17     printf("Vehicle inserted = %d", vehicle);
18 }
19
20 int main() {
21     enqueue(40);
22     return 0;
23 }
```

Test Input

40
0
2

Output

Vehicle inserted = 40

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 3;
7
8 void push(int data) {
9     if (top == MAX - 1) {
10         printf("Overflow");
11         return;
12     }
13
14     stack[++top] = data;
15     printf("Inserted element = %d", data);
16 }
17
18 int main() {
19     push(50);
20     return 0;
21 }
```

Test Input

4
60

Output

Inserted element = 50

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 3;
7
8 void push(int data) {
9     if (top == MAX - 1) {
10         printf("Overflow");
11         return;
12     }
13
14     stack[++top] = data;
15     printf("Inserted element = %d", data);
16 }
17
18 int main() {
19     push(50);
20     return 0;
21 }
```

Test Input

4
60

Output

Inserted element = 50

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Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11    return stack[top--];
12 }
13
14 int main() {
15     int item = pop();
16
17     if (item != -1)
18         printf("Popped element = %d", item);
19
20     return 0;
21 }

```

Test Input

Empty
-1

Output

Popped element = 30

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Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *prev, *next;
7 };
8
9 void deleteNode(struct Node **head, struct Node *del) {
10    if (del == NULL)
11        return;
12
13    if (*head == del)
14        *head = del->next;
15
16    if (del->prev != NULL)
17        del->prev->next = del->next;

```

Test Input

10 <-> 20
20

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192525345
RA_6a7982a1d68085.33373732\main.c
: In function 'main':
C:\Windows\TEMP\sandbox_192525345
RA_6a7982a1d68085.33373732\main.c

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int rear = 2;
7
8 void enqueue(int data) {
9     if (rear == MAX - 1) {
10         printf("Queue Overflow\n");
11         return;
12     }
13     queue[++rear] = data;
14 }
15
16
17 int main() {
18     enqueue(40);
19     printf("Inserted element = %d", queue[rear]);
20 }
```

Test Input

2
40

Output

Inserted element = 40

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Q3
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int deque[MAX];
6 int front = 2, rear = 4;
7
8 void insertFront(int data) {
9     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
10         printf("Deque Overflow");
11         return;
12     }
13
14     if (front == 0)
15         front = MAX - 1;
16     else
17         front = front - 1;
18
19     deque[front] = data;
20     printf("Inserted element = %d", data);
21 }
22
23 int main() {
```

Test Input

2
4
10

Output

Inserted element = 10

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Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l > r)
5         return -1;
6
7     int mid = l + (r - l) / 2;
8
9     if (arr[mid] == x)
10        return mid;
11
12    if (arr[mid] > x)
13        return binarySearch(arr, l, mid - 1, x);
14
15    return binarySearch(arr, mid + 1, r, x);
16 }
17
18 int main() {
19     int arr[] = {10, 20, 30, 40, 50};
20     int n = 5, x = 40;
21
22     int result = binarySearch(arr, 0, n - 1, x);
```

Test Input

10 20 30 40 50
40

Output

Element found at index 3

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Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX] = {10, 20, 30};
6 int front = 0, rear = 2;
7
8 int dequeue() {
9     if (front == -1) {
10        printf("Queue Underflow\n");
11        return -1;
12    }
13
14    int data = queue[front];
15
16    if (front == rear) {
17        front = rear = -1;
18    } else {
19        front = (front + 1) % MAX;
20    }
21
22    return data;
```

Test Input

{10, 20, 30}
0
2

Output

Deleted element = 10

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Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[10], top = -1;
4
5 void push(int x) {
6     stack[++top] = x;
7 }
8
9 int pop() {
10    return stack[top--];
11 }
12
13 int main() {
14    push(2);
15    push(3);
16
17    int op2 = pop();
18    int op1 = pop();
19
20    int result = op1 + op2;
```

Test Input

23+

Output

Result = 5

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